



2024 - Baulkham year 9 past papers

Year 9 Math (Baulkham Hills High School)



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Year 9 Common Test – Term 2 2024

TIME: 60 minutes + 5 minutes reading time

Name:

Teacher:

Directions

- Full working should be shown in every question.
- Marks may be deducted for careless or badly arranged work.
- Use non-erasable black or blue pen only (not pencils) to write your solutions.
- No liquid paper / correction tape is to be used.
If a correction is to be made, one line is to be ruled through the incorrect answer.
- The diagrams are not drawn to scale.
- Calculators are allowed in this exam.

(Teacher use only)

Marking Grid

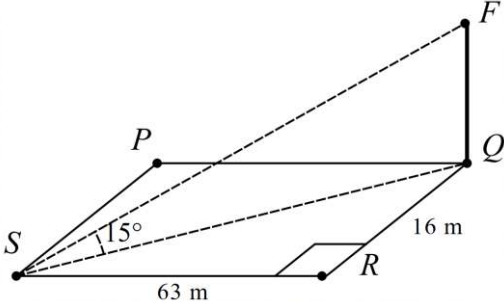
	Financial Mathematics	Trigonometry	Algebra	Coordinate Geometry	Totals
Page 2	3 /1		1,2,4 /12		/ 13
Page 3	8,10 /5		5,6,9 /5	7 /2	/ 12
Page 4	11 /2	12 /4		13 /5	/ 11
Page 5	16 /2	17 /3	14 /2	15 /1	/ 8
Page 6		19,20 /6	18 /2		/ 8
Page 7				21,22 /6	/ 6
Multiple choice	2 /1	1,5 /2	7,8,9 /3	3,4,6,10 /4	/ 10
Totals	/ 11	/ 15	/ 24	/ 18	/ 68

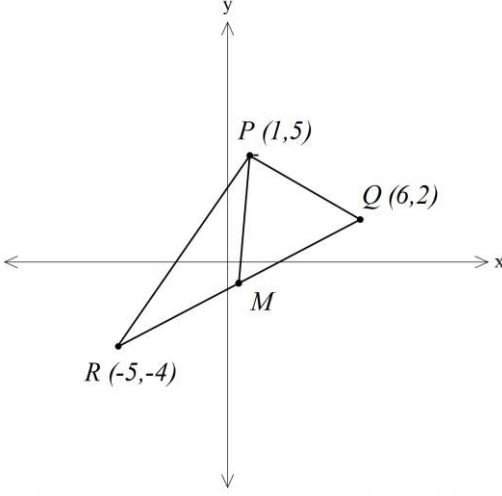
Part A – Written response

1) Simplify the following:	
a) $3a^2 \times (-3b) \times 2a^3 \times 5b^2$	1
b) $\frac{3}{2x} - \frac{5}{3x}$	1
c) $\frac{10x^3 - 1000x}{x - 10}$	2
2) Fully factorise the following:	
a) $7pq^2 - 14p^5q^5 + 21p^3q^2$	1
b) $5x^2 + 37x + 14$	2

c) $-2pq^2 + 4p^2q - 4p^2q^5 + 8p^3q^4$	2
3) A toy is sold to make a profit of 60%. What was its cost price if the toy was sold for \$120?	1
4) Simplify the following.	3
$\frac{p+5}{p-2} \div \frac{p^2+3p-10}{p^2-4p+4}$	

5)	Simplify the following.	2
	$\frac{64x^3 - 27y^3}{8x - 6y}$	
6)	Make x the subject of the following equation.	2
	$a^2x^2 - b^2 = b^2c^2$	
7)	The points $A(2,3)$, $B(4,0)$ and $C(k-3,-3)$ are collinear. Find the value of k .	2
8)	Ben earns \$53.20 per hour for a 37-hour week as a plumber. Overtime is paid at time and a half rate for the first 5 hours and double time thereafter. Calculate Ben's pay for a week when he works for 48 hours.	2
9)	Factorise $64x^2 + 80xy + 25y^2$.	1
10)	Jack bought a laptop for his business for \$2500, which depreciates at a rate of 40% p.a.	
	a) What is the value of the laptop at the end of the fourth year?	1
	b) Calculate the total percentage loss of the value of the laptop at the end of the fourth year.	2

11)	Fred invested an amount of money which accumulated to \$12,000 after 6 years. Interest was paid at a rate of 9% per annum, compounded monthly. How much did Fred invest? (Give your answer to the nearest dollar)	2
12)	A lamp post is erected at corner Q of a rectangular courtyard $PQRS$ with side lengths 63 m and 16 m as shown. The angle of elevation of the top of the lamp post from S is 15°. NOT TO SCALE  a) Calculate the height in metres, of the lamp post to two decimal places. b) Calculate the angle of elevation of the top of the lamp post from R to the nearest degree.	2

13)	The points $P(1,5)$, $Q(6,2)$ and $R(-5,-4)$ form the vertices of a $\triangle PQR$, where M is the midpoint of QR as shown in the diagram below. NOT TO SCALE  a) Find the coordinates of M. b) Find the equation of PM. c) Calculate the length of RQ to two significant figures.	1 2 2
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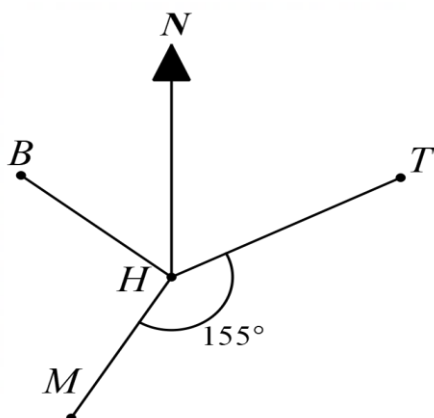
- 18) A freight train F, leaves Central Station towards Orange at 40 km/h. Two hours later, a passenger boards another train P travelling at 60 km/h heading towards Orange as well. By forming an equation and solving it, find how long it will take for the passenger train P to overtake the freight train F.
(let x = time in hours for the passenger train)

2

- 19) The diagram below shows the surrounding buildings relative to a hospital (H), where the mall (M) is southwest of H , the bus stop (B) is northwest of H and the train station (T) is marked as shown, where $\angle THM$ is 155° . What is the bearing of T from H?

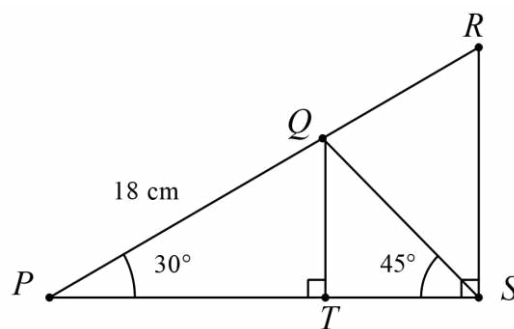
2

NOT TO SCALE



- 20) Find the exact value of QR if $PQ = 18$ cm.

4

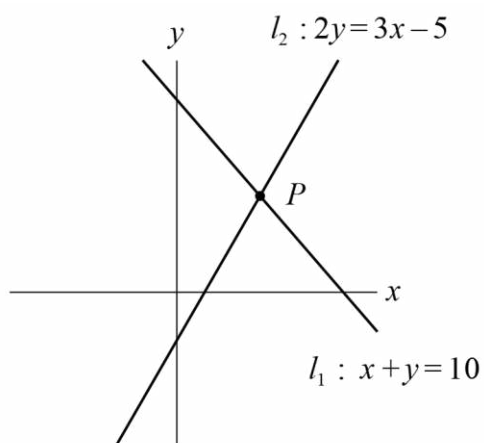


NOT TO SCALE

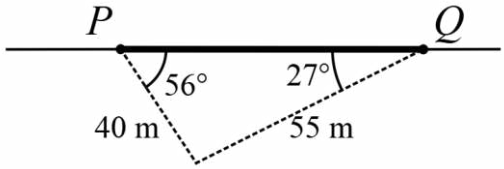
21) Find the equation of the straight line passing through the point of intersection of $3x + 4y - 20 = 0$ and $2x - 2y - 4 = 0$ and perpendicular to the straight line $6x + 3y - 7 = 0$. 3

22) Two straight lines l_1 and l_2 where, 3
 $l_1 : x + y = 10$ and
 $l_2 : 2y = 3x - 5$
 intersect at the point P as shown in the diagram.

NOT TO SCALE



Find the area enclosed by l_1, l_2 and the y -axis.

Part B - MULTIPLE CHOICE (1 mark each) <i>Circle the correct answer.</i>	
1)	If $\sin \theta = 0.632$, what is the value of θ ? (A) $39^{\circ}11'$ (B) 39.19° (C) $39^{\circ}12'$ (D) $39^{\circ}20'$
2)	How much interest is paid on a monthly balance of \$715 with a simple interest rate of 7.2% p.a.? (A) \$4.29 (B) \$51.48 (C) \$429 (D) \$5.15
3)	What is the equation of the line that intercepts the y-axis at -1 and has a gradient equivalent to 3? (A) $2y = 3x - 1$ (B) $-6x + 2y + 2 = 0$ (C) $2y = -3x + 1$ (D) $y = 3x + 1$
4)	Which of the following points does not lie on the graph of $y = -2x + 7$? (A) (4, -1) (B) (-1, 9) (C) (-2, 11) (D) (-3, 1)
5)	A foot bridge is constructed as below. What is the length of PQ?  (A) $95 \sin 97^{\circ}$ (B) $40 \sin 56^{\circ} + 55 \sin 27^{\circ}$ (C) $40 \cos 56^{\circ} + 55 \cos 27^{\circ}$ (D) $95 \cos 97^{\circ}$

6)	If (4, 2) is the mid-point of the line joining (m, 4) and (3, n), then what is the value of (m - n) ? (A) 5 (B) 6 (C) 7 (D) 0
7)	A 30-carat diamond was broken into two pieces. The two pieces were placed on a balance, one on each side. To maintain balance, an additional 12 carat weight plate was required. What is the weight of the larger piece? (A) 18 carats (B) 21 carats (C) 24 carats (D) 26 carats
8)	Which of the following shows the correct equation where x is the subject of $\sqrt[3]{\frac{x}{p}} = m$? (A) $x = p^3 m$ (B) $x = \frac{m^3}{p^3}$ (C) $x = (pm)^3$ (D) $x = pm^3$
9)	Which of the following expressions has the factors (x + 3) and (x - 5) ? (A) $x^2 - 2x - 15$ (B) $x^2 + 2x - 15$ (C) $x^2 + 2x + 15$ (D) $x^2 - 2x + 15$
10)	For what value of k are the linear equations $y = 2x - 8$ and $y - kx - 7 = 0$ parallel to each other? (A) 7 (B) 2 (C) -7 (D) -2

End of test

Extra Working Space

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